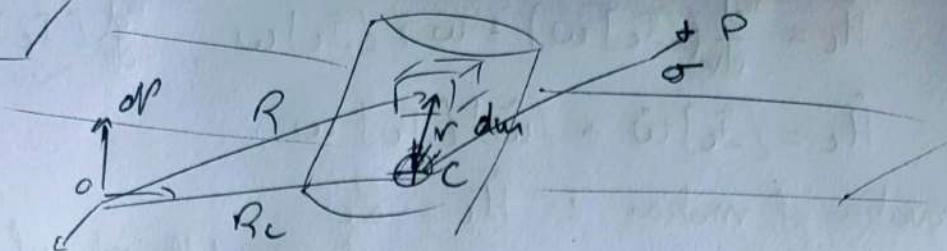


PROBLEM 1



- (A) By definition $H_P = \int_B (\sigma \times \dot{\sigma}) dm$ with $\sigma = R - R_P$.
Specific case where we consider $P = C$

$$\sigma = R - R_C \rightarrow \underbrace{R - R_C}_{\sigma} = r$$

$$H_C = \int_B (r \times \dot{r}) dm$$

- (B) $\dot{r} = \frac{d}{dt} r + \omega \times r$ (transport theorem)

assuming rigid body we have $\frac{d}{dt} r = 0$.

$$\dot{r} = \omega \times r$$

$$\frac{d}{dt} H_C = \dot{H}_C = \frac{d}{dt} H_C + \omega \times H_C$$

$$H_C = \int_B (r \times (\omega \times r)) dm$$

$$H_C = \int_B r \times (-r \times \omega) dm$$

$$a \times b = -b \times a$$

Using the skew symmetric operator we have $r \times \omega = [\tilde{r}] \omega$

$$H_C = \int_B [\tilde{r}] (-[\tilde{r}] \omega) dm = - \int_B [\tilde{r}] [\tilde{r}] \omega dm$$

Since we consider a rigid body we have ω constant in B :

$$H_C = \left(- \int_B [\tilde{r}] [\tilde{r}] dm \right) \omega = \underline{[\mathbb{I}_C]} \omega$$

$\stackrel{\Delta}{=} [\mathbb{I}_C]$

Finally: $\dot{H}_c = \frac{Bd}{dt} ([I_c] \omega) + \omega \times [I_c] \omega$, $\frac{Bd}{dt} [I_c] = 0$ (rigid)

$$\dot{H}_c = [I_c] \dot{\omega} + [\tilde{\omega}] [I_c] \omega$$

Euler equation of motion: $\dot{H}_c = \mathcal{L}_c$
 $\mathcal{L}_c$ = sum of the external torques acting on the spacecraft.

$$\rightarrow \mathcal{L}_c = [I_c] \dot{\omega} + [\tilde{\omega}] [I_c] \omega$$

(c) By definition: $T = \frac{1}{2} \int_B \dot{R} \cdot \dot{R} \, dm$

$$R = R_c + r \rightarrow \dot{R} = \dot{R}_c + \dot{r}$$

$$\hookrightarrow T = \frac{1}{2} \int_B (\dot{R}_c + \dot{r}) \cdot (\dot{R}_c + \dot{r}) \, dm$$

$$T = \frac{1}{2} \int_B [(\dot{R}_c \cdot \dot{R}_c) + (\dot{r} \cdot \dot{R}_c) + (\dot{r} \cdot \dot{R}_c) + (\dot{r} \cdot \dot{r})] \, dm$$

$$T = \frac{1}{2} \int_B (\dot{R}_c \cdot \dot{R}_c) \, dm + \underbrace{\int_B (\dot{r} \cdot \dot{R}_c) \, dm}_{\dot{R}_c \cdot \int_B \dot{r} \, dm = 0 \text{ (rigid body)}} + \frac{1}{2} \int_B (\dot{r} \cdot \dot{r}) \, dm$$

$$T = \frac{1}{2} M (\dot{R}_c \cdot \dot{R}_c) + \frac{1}{2} \int_B (\dot{r} \cdot \dot{r}) \, dm = T_{\text{translat}} + T_{\text{rot}}$$

$$T_{\text{rot}} = \frac{1}{2} \int_B (\omega \times r) \cdot (\omega \times r) \, dm$$

$(a \times b) \cdot c = a \cdot (b \times c)$
 $(\omega \times r) \cdot (\omega \times r) = \omega \cdot (r \times (\omega \times r))$

$$T_{\text{rot}} = \frac{1}{2} \int_B \omega \cdot (r \times (\omega \times r)) \, dm = -\frac{1}{2} \int_B \omega \cdot [\tilde{r}] [\tilde{r}] \omega \, dm$$

$$T_{\text{rot}} = \frac{1}{2} \omega^T \underbrace{\int_B -[\tilde{r}] [\tilde{r}] \omega \, dm}_{\text{RBR}}$$

$$|T_{\text{rot}} = \frac{1}{2} \omega^T [I_c] \omega|$$

$$c) \quad |T = \frac{1}{2} [M (\dot{R}_c \cdot \dot{R}_c) + \omega^T [I_c] \omega]|$$